RAKSHAM · AI CYBERSECURITY HACKATHON · ROUND 1 · PROBLEM 02

Satark

सतर्क

The scam shield that steps in **before** the UPI payment, not after.

It joins what a scammer says with the payment being attempted, on the user's own phone, and answers with a specific, graded alert.

On-device · Never auto-blocks · One-tap report



TEAM

WINNER

Moksh Kulshrestha (team leader) · Mitalika Khosla · Piyush Wahi · Shivam Thakur

The harm lands at one moment: a real person, under pressure, approves a payment the bank sees as normal

1 Unknown caller



claims a parcel or SIM links you to a crime

2 Fake authority



forged warrant, "CBI" on video

3 Isolation



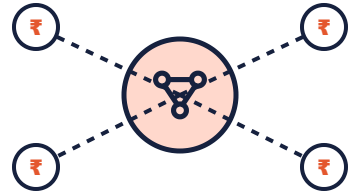
stay on the call, tell no one

4 The ask



move savings to a "safe" account

5 Mules

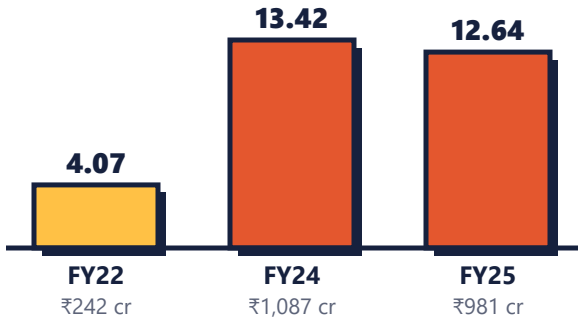


funds layered within minutes

WHO WE PROTECT

"Meena", 62 (illustrative). Pays bills by UPI, never sent a large sum to a new payee, often alone when the call comes.

UPI FRAUD CASES, LAKH



FY22, FY24, FY25 as reported by the Finance Ministry in Lok Sabha; amounts in Rs crore.

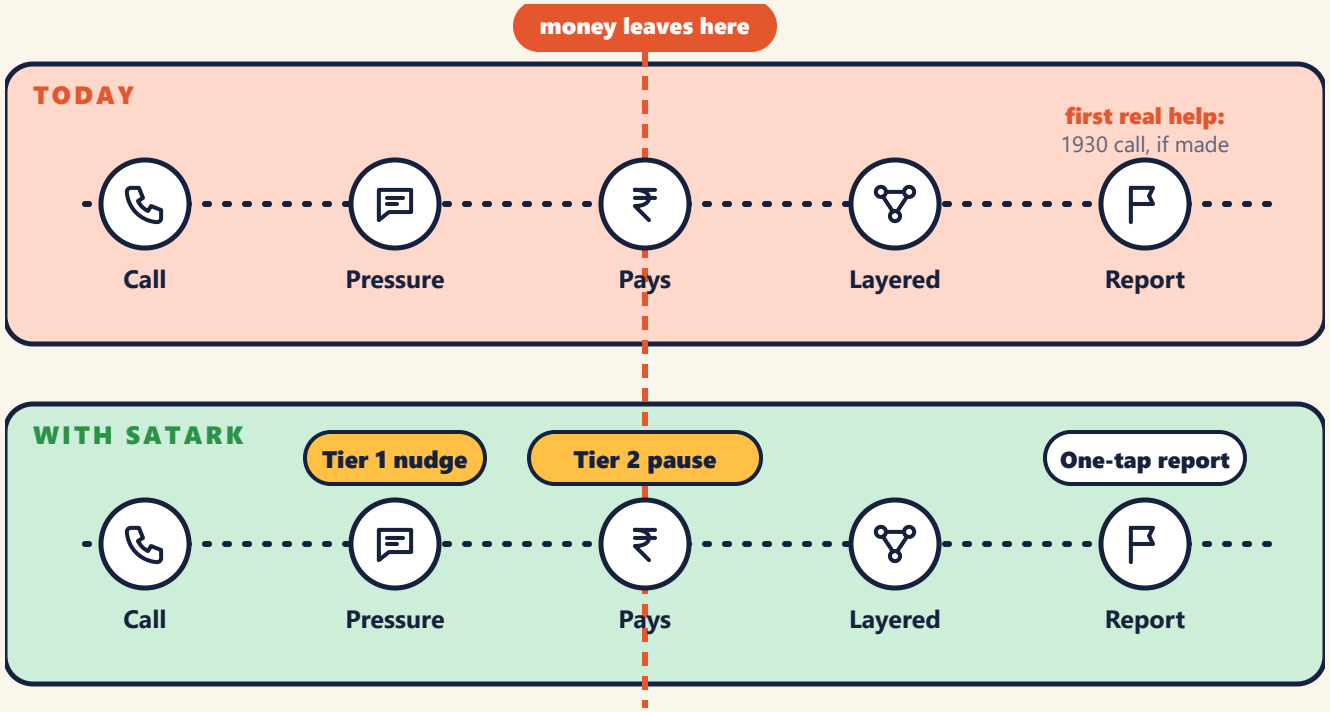
51%
of UPI-fraud victims never filed a complaint (LocalCircles, 2025)

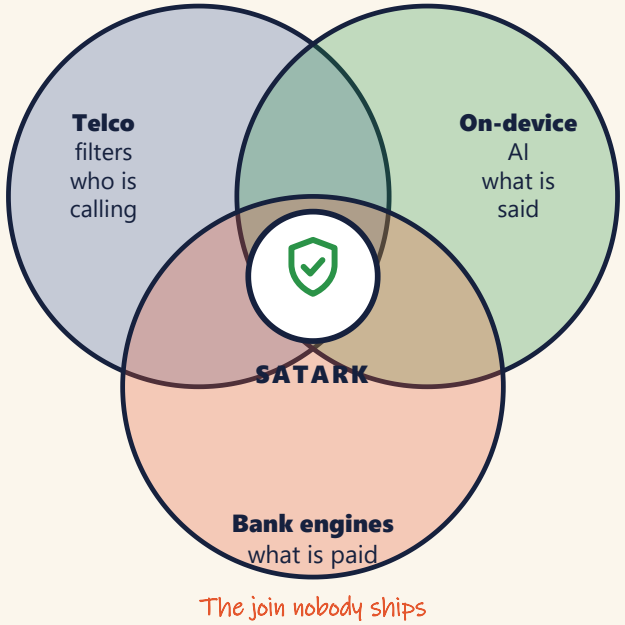
47%
of Indian adults met someone hit by an AI voice scam (McAfee 2023)

THE PATTERN WE DETECT

unknown caller + authority and urgency + isolation + ₹ first-time large payee

Same call, same scammer. Satark moves help from after the payment to before it





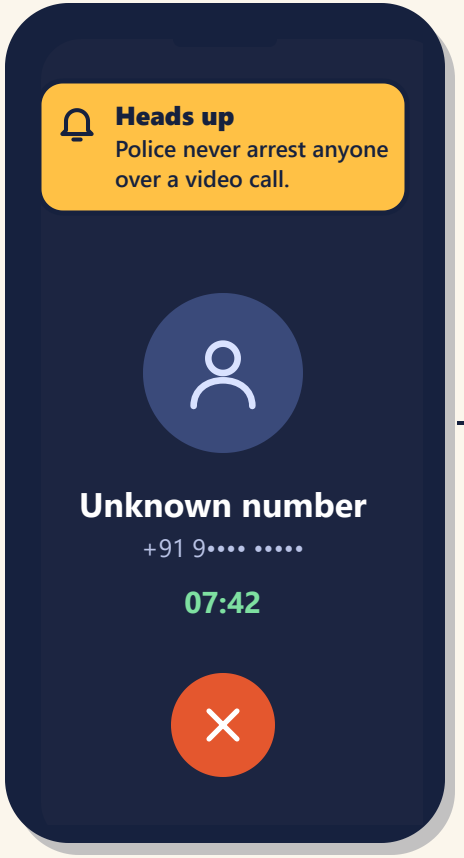
Each layer sees one slice. Satark joins all three, on the phone.

Valid on paper. The bank sees a real customer approving a payment, so limits and generic prompts are easily coached past.

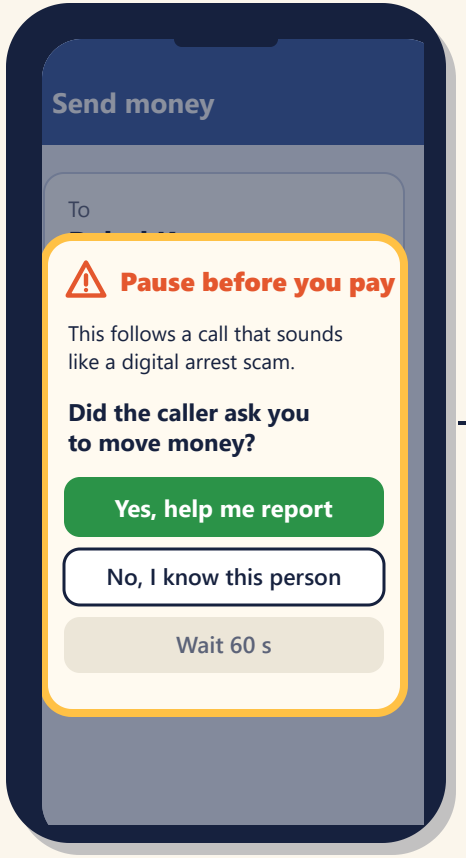
Minutes, not hours. Funds are layered through mule accounts almost at once; freezes depend on a fast 1930 call.

Satark's bet. Act at the moment of decision, with a specific message, and make reporting one tap. *(To our knowledge, from public sources; see Appendix B.)*

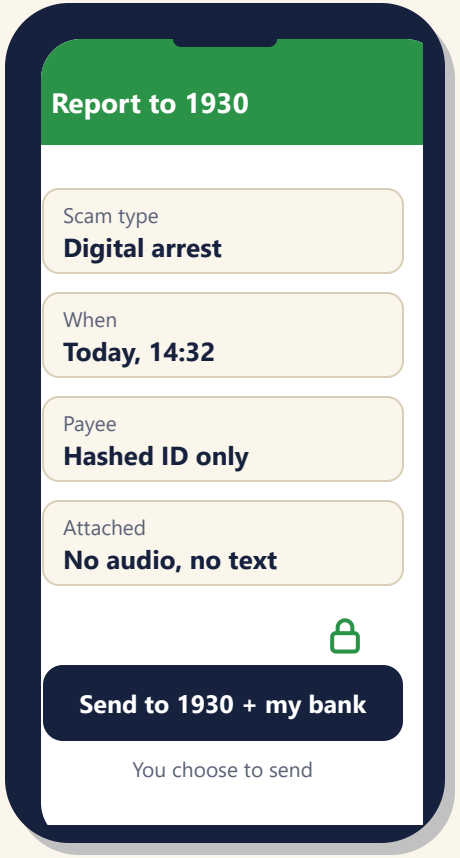
From a suspicious call to a safe decision in three taps



1 Call
Tier 1 nudge names the scam



2 Payment
Tier 2 pause + one question



3 Report
Prefilled 1930 report, user sends

ON THE PHONE

Call or message text (in memory), contact metadata and the payment event become scores. Opt-in per session, visible indicator, one-tap off.

WHAT THE USER SEES

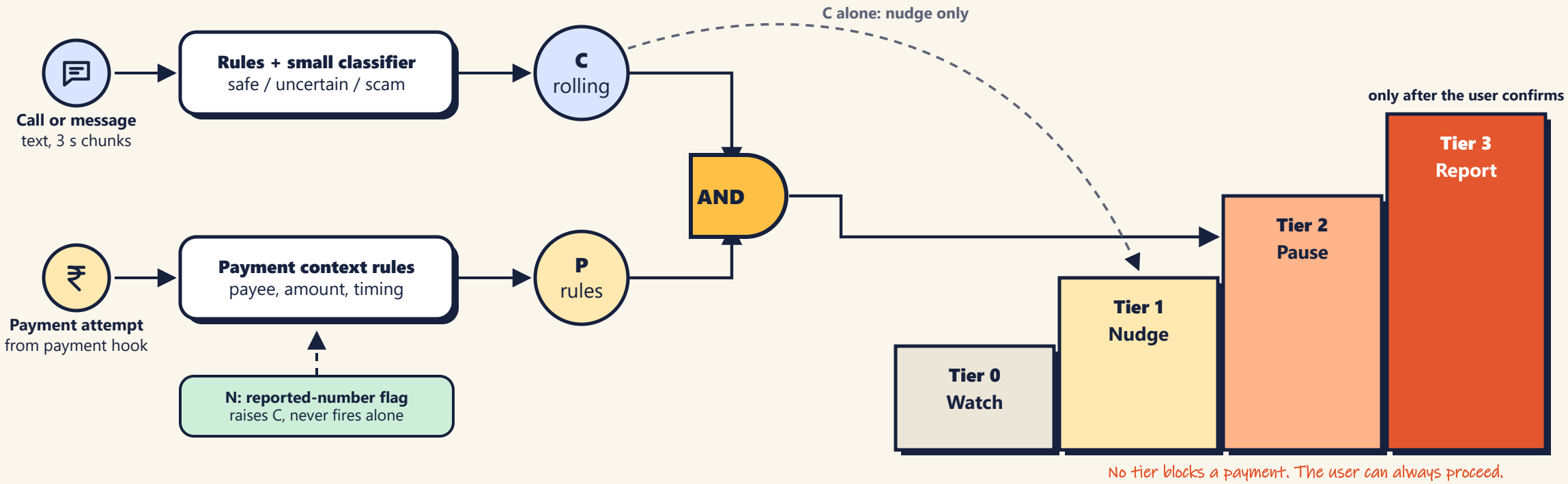
A short message naming the scam and one question. Never a raw score.

WHAT LEAVES

Only a user-approved report pack: scam type, tier, times, features. No audio, no transcript.

three taps: nudge, pause, report

A pause needs two independent signals to agree; one alone only earns a nudge



Layer 1 · Rules. Authority words, "tell no one", new payee, amount vs own median, timing.

Layer 2 · Small classifier. DistilBERT/RoBERTa-class, fine-tuned on ICFD-31k chunks plus our Hinglish scenarios.

Layer 3 · Uncertain only. Ask the user a question; optional on-device LLM, never cloud.

Public and synthetic data only, and a test for every assumption

31,000+
Indian call transcripts (ICFD-31k)

1.1 M
3-second chunks with streaming labels

10 + 5
fraud types, 5 held out as unseen

0
real customer or bank records used

HOW WE VALIDATE

- **Streaming test:** recall and time-to-detect, including unseen fraud types
- **Joint scenarios:** scam and benign journeys: tier-2 false positives, alerts per user per week
- **Language slices:** Hindi, Hinglish, English reported separately
- **Message test:** volunteers compare alert wording with a generic warning

Demo-scale results, reported as such.

ASSUMPTIONS, HONESTLY

A1 Synthetic scripts resemble real scams. test: check on small real public sets

A2 Call text needs consent and a partner (OS, telco or app). demo: user-started transcription

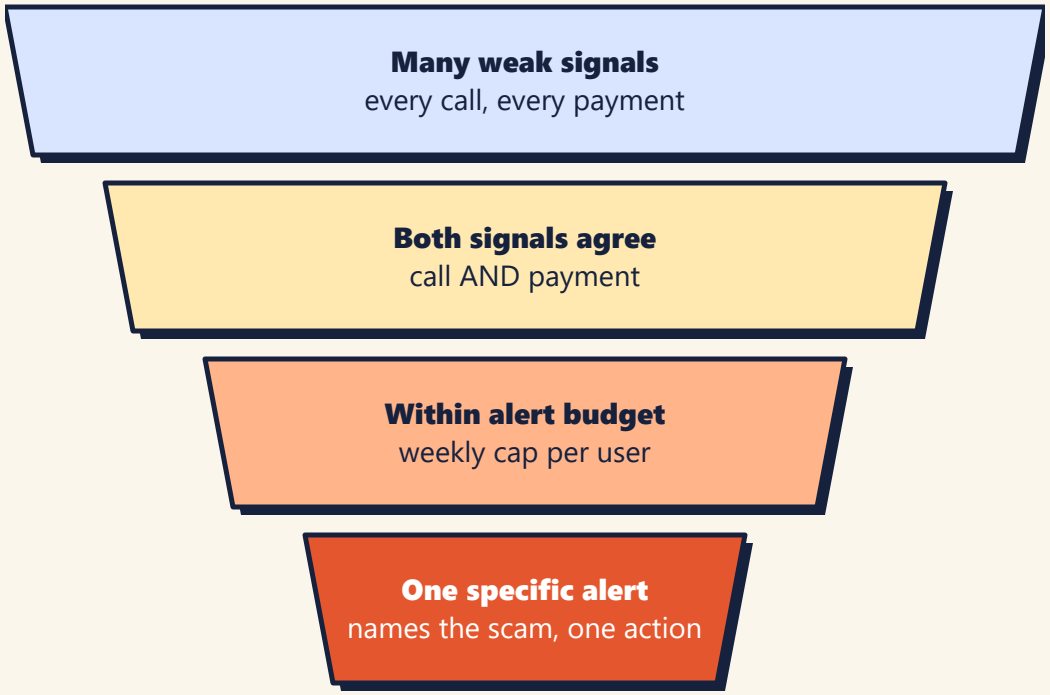
A3 A payment hook can read payee and amount. demo: mock screen

A4 People act on a specific question. test: message study

DATA WE USE (ALL PUBLIC OR SYNTHETIC)

ICFD-31k · Hinglish scam text · Indian multilingual set · Smishing IMC'25 · Smishtank · IBM AML-Data · PaySim · our UPI scenario generator

Fewer, sharper alerts, and a recovery path for every mistake



illustrative: shape of the filter, not measured numbers

Wording is specific and changes between alerts. A weekly budget and repeated dismissals lower noise. Generic warnings fail: only about a third of UK adults read their bank's (Opinium, 2024).

FALSE POSITIVE

Pause on a legitimate payment after a stressful call.

→ "I know this person" proceeds at once and lowers that sender's weight. Nothing is blocked.

FALSE NEGATIVE

Scam not flagged, user pays.

→ "I was scammed" starts the 1930 report now; the pattern joins the retraining set.

UNCERTAIN

Signals do not agree.

→ One neutral question, never an accusation; no answer means no escalation.

LANGUAGE BIAS

Weaker accuracy in some languages or speech styles.

→ Per-language results and thresholds; the gap is reported.

No automatic block or accusation on one weak signal. The user always decides; only a human at the bank or I4C can freeze funds.

Raw content never leaves the phone, and the tool is hard to turn into a weapon



Never collected: audio recordings, full transcripts, contact list, bank credentials, UPI PIN, OTP

Coached victim
varying wording, two-signal gate

Evasion
uncertain tier, retraining, unseen-type tests

Poisoned list
signed feed, corroboration, rate limits

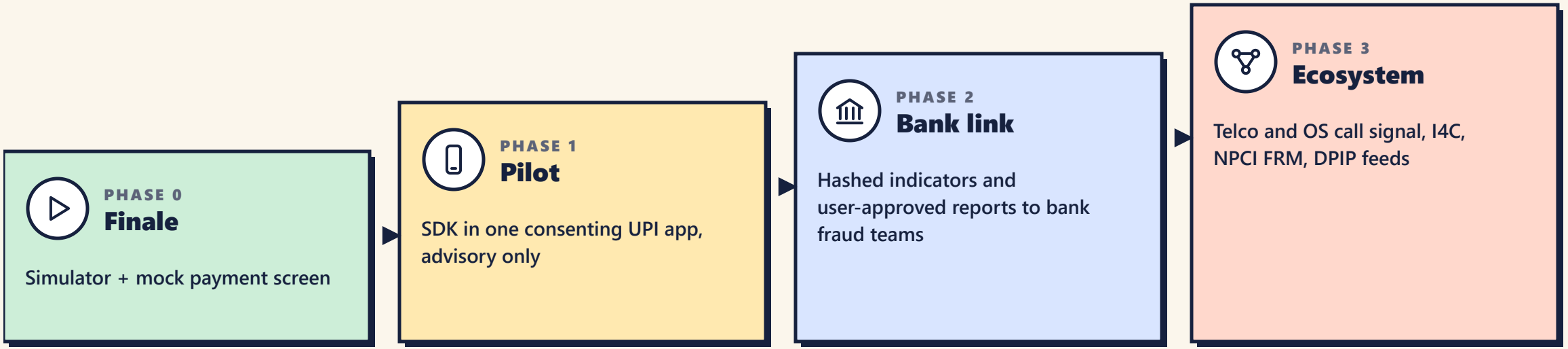
Stolen evidence
features only, encrypted, auto-expiry

Stalkerware use
visible, per-session, no remote access

Alert flooding
per-sender caps, alert budget

Designed around DPDP Act 2023 principles (consent, purpose limitation, minimisation; legal review needed) and RBI FREE-AI ideas: human oversight, explainability.

Start as an advisory layer in one UPI app, then connect to bank and national fraud intelligence



Phases 1 to 3 need partners we do not have yet: a roadmap, not a claim.

Owner. The UPI app or bank shipping the SDK; I4C and bank fraud teams own human review.

Latency. Scoring runs on device per 3 s chunk; the payment check adds one rule lookup. Target measured at the finale.

Integration. Payment-screen SDK hook; prefilled 1930 report the user submits; no change to UPI rails.

A narrow MVP: one scam, end to end, with measured alert quality



flags = checkpoints: each ends in a runnable demo step and a logged metric

▷ SHOWN AT HOUR 48

- Scripted digital-arrest call: nudge, then payment pause
- "Scam" answer gives the 1930 report pack
- Benign call to a known payee: no alert
- Metrics: detection time, false positives, alerts per user

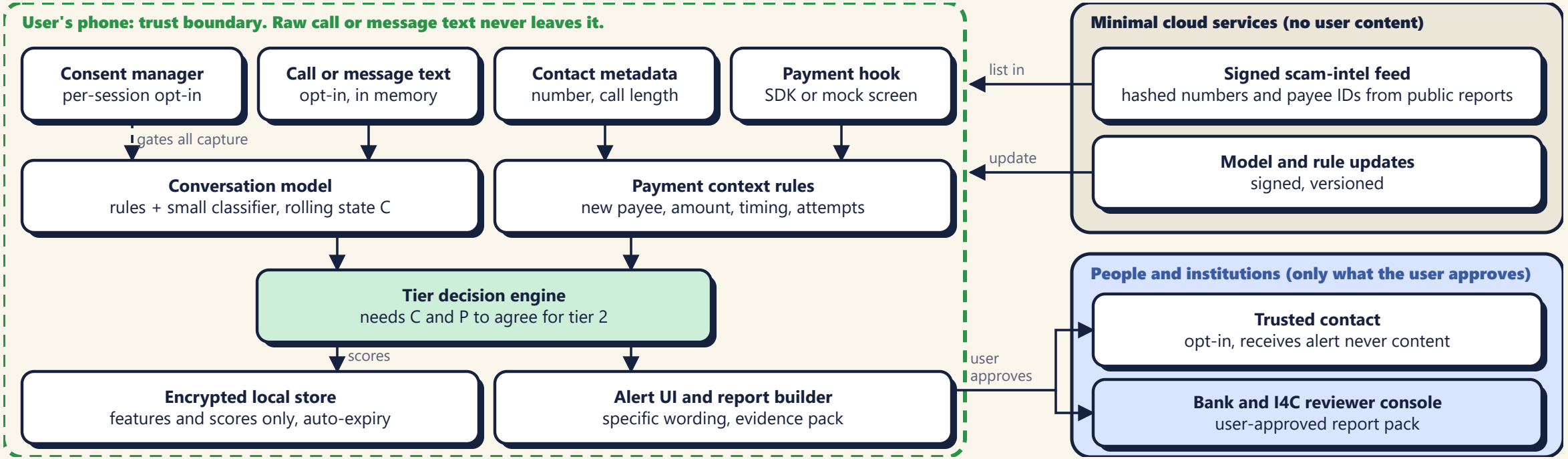
FOUR ROLES

- ML and data
- Rules and backend
- App and UI
- Safety, wording and demo

FALLBACKS

- No live call audio: scripted transcripts
- Model too slow: distilled or rules-only
- No Android build: web simulator
- LLM layer dropped first

System diagram: scoring stays on the phone, the cloud holds no user content

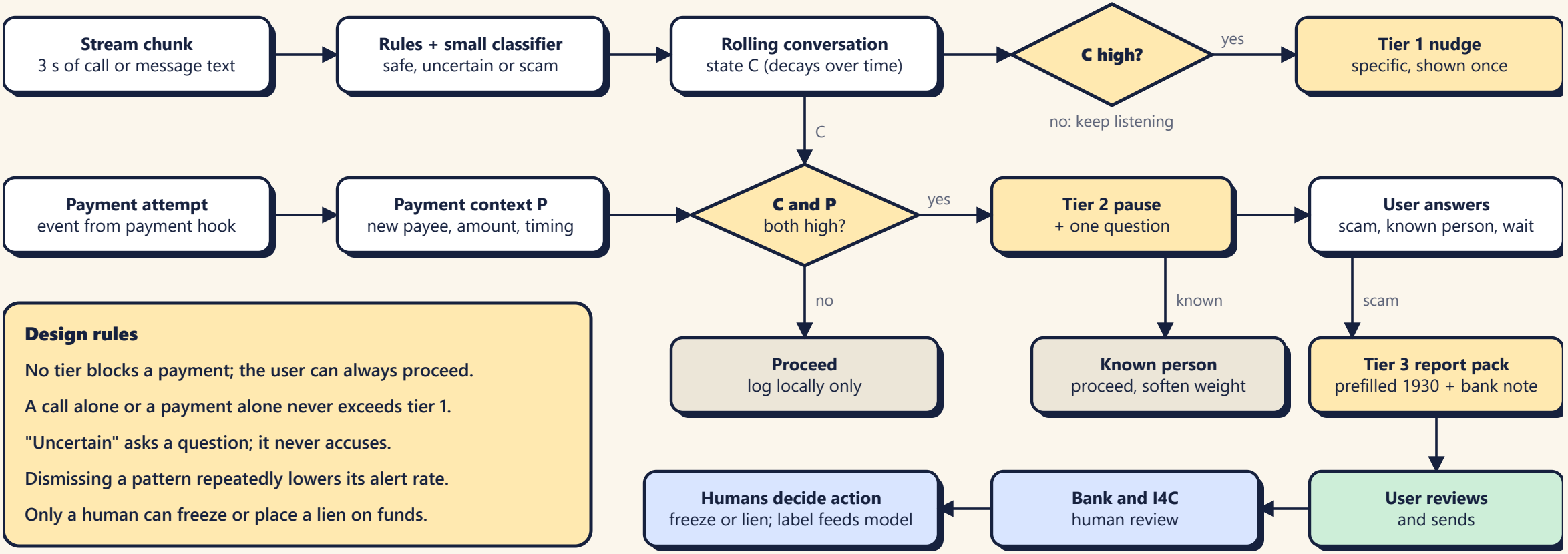


Inputs and consent. Text (opt-in per session), contact metadata, payment event. Training data public or synthetic only (Appendix A).

Components. Rules, small DistilBERT/RoBERTa-class classifier, payment rules, tier engine, encrypted store, report builder.

Boundaries. Cloud is inbound-only (signed list, updates). Outbound only on user approval, and never content.

Decision flow: two signals must agree, the user stays in control, humans make the high-impact call



Design rules

No tier blocks a payment; the user can always proceed.

A call alone or a payment alone never exceeds tier 1.

"Uncertain" asks a question; it never accuses.

Dismissing a pattern repeatedly lowers its alert rate.

Only a human can freeze or place a lien on funds.

Alerts. Tier 1 nudge, tier 2 pause with one question; scam-specific wording, no score shown.

Evidence. Report pack: scam type, tier, times, features, hashed payee ID. No audio or transcript.

Human review. Bank fraud team and I4C review; only they decide on a freeze or lien. Their verdict becomes a label.

Mitigations, performance assumptions and how we will prove them

RISK	MITIGATION
Model theft or tampering	Signed model bundles; integrity check at load; no secrets in the model
Evidence pack exposure	Features only; encrypted; auto-expiry; user-held export key
Scam-list poisoning	Signed feed; corroboration across sources; human curation
Evasion, prompt injection in transcript	Classifier treats text as data; "uncertain" tier; unseen-type tests; no model output executes actions
Privacy abuse (stalkerware)	Visible indicator, per-session opt-in, no remote access, trusted contact never gets content
Unauthorised evidence access	Reviewer console behind bank or I4C authentication; access logged; shared only on user action
Alert flooding	Per-sender caps, alert budget

PERFORMANCE ASSUMPTIONS (TARGETS, TO BE MEASURED)

- Chunk scoring on a mid-range phone well within the 3 s interval
- Payment check adds one rule evaluation before the PIN screen
- Baseline: ICFD-31k streaming RoBERTa F1 0.956, 30 s median detection (authors, synthetic)
- Target: tier 2 fires before the first large transfer in scripted scenarios

VALIDATION

- Held-out and unseen fraud types
- Joint call-plus-payment scenarios
- Per-language slices
- Red-team round: paraphrased and coached scripts
- Small alert-wording test

WHAT WE WILL MEASURE AND REPORT AT THE FINALE

time to detect · **tier-2 false positives** · **alerts per user per week** · **per-language gap** · **unseen-type recall** · **on-device latency**

APPENDIX A · DISCLOSURE

AI-generated content, datasets, models and third-party assets, with licence status

ITEM	TYPE	USE	LICENCE OR STATUS
Text, diagrams and illustrations in this PDF	AI-assisted	Research summary, drafting and SVG artwork produced with Claude (Anthropic); reviewed and edited by the team	Team-owned; no stock or third-party images
ICFD-31k	Dataset	Conversation model training and streaming evaluation	Code MIT; data research-use only, no scam-script generation (we hand-write templates)
Hinglish and English Financial Scam Text	Dataset	Seed phrases; English part derives from UCI SMS Spam	CC BY 4.0
Indian Multilingual Scam Message Dataset (120 rows)	Dataset	Evaluation sanity check	Apache-2.0
Smishing Dataset IMC'25	Dataset	Lure taxonomy, message structure	CC BY 4.0
Smishtank	Dataset	URL and domain features	CC BY-NC-SA 4.0 (non-commercial; research use only)
IBM AML-Data	Synthetic dataset	Mule and fund-flow pattern features	Data CDLA-Sharing-1.0; repository Apache-2.0
PaySim	Synthetic dataset	Transaction-pattern reference	Licence to be confirmed on Kaggle page before use
Our synthetic UPI scenario generator	Team-built	Joins scam scripts to payment sequences, plus benign controls	Team-owned; hand-written templates
DistilBERT or RoBERTa base models	Pretrained model	Fine-tuned chunk classifier	Apache-2.0 (DistilBERT); MIT (RoBERTa); confirm model card at build time
PyTorch, Transformers, scikit-learn, FastAPI	Open-source libraries	Training, inference, API	Open-source licences, listed in the prototype repository
Fonts	System fonts	Georgia, Segoe UI, Bahnschrift, Ink Free as installed on the build machine	Not redistributed

No confidential bank, customer or platform data was used or will be used. All scenarios, transcripts and transactions in the prototype are synthetic or public.

APPENDIX B · SOURCES

Where the facts and baselines in this document come from

STATISTICS AND CONTEXT

- Finance Ministry reply to Lok Sabha on UPI fraud cases and losses, FY2021-22 to FY2024-25 (reported by BOOM and others).
- LocalCircles survey on UPI fraud, 2025 (reported by Business Standard).
- McAfee, "Beware the Artificial Impostor", 2023 (reported by BOOM).
- BOOM and CYFIRMA reports on malicious APK and KYCShadow scams.
- RBI FREE-AI committee report, Aug 2025; RBI Annual Report 2024-25.
- RBIH MuleHunter.AI and Digital Payments Intelligence Platform (RBI and press statements); NPCI fraud risk management service.
- Supreme Court proceedings on digital-arrest scams and mule-account SOP (Bar and Bench).
- Airtel AI spam detection (GSMA case study); Google Safety Charter for India and on-device scam detection; PhonePe Protect; Paytm Payments Bank AI security.

RESEARCH AND DATASETS

- Ahuja, Prateek, Singh. ICFD-31k: Dataset and Benchmark for Real-Time Conversational Fraud Detection. IJCAI-ECAI 2026.
- "It Warned Me Just at the Right Moment": LLM-based real-time detection of phone scams. arXiv 2502.03964.
- Explainable Agentic System for Detection of Conversational Scams with Summary-Based Memory. arXiv 2607.11707.
- Detection, Attribution, Narration: explainable money-mule identification. arXiv 2607.17586.
- GraphSAGE on large non-bipartite transaction networks. arXiv 2509.12255.
- Altman et al. AMLworld synthetic transactions. arXiv 2306.16424.
- Smishing Dataset I (Smishtank). arXiv 2402.18430; Smishing Dataset IMC 2025.
- Egelman et al. and SOUPS 2025 poster on phishing warnings; Opinium and Tunic Pay UK warning survey, 2024.
- Repositories studied: Rupali-2507/MULE_HUNTER, SPELLAILab/ICFD-31k, IBM/AMLSim, IBM/AML-Data, ipridem/smishguard.

Figures are quoted as reported by the cited sources and may be revised. The PDF is self-contained; no live links are needed.